MIL-B-18558A(SHIPS)
13 December 1960
SUPERSEDING
MIL-B-18558 (SHIPS)
24 May 1955

MILITARY SPECIFICATION

BEARING UNITS, PLAIN, PROPULSION, LINE SHAFT

1. SCOPE

1.1 Scope. - This specification covers plain propulsion line shaft bearing units for Naval ships.

1.2 Classification. - Propulsion line shaft bearing units shall be of the following types, as specified (see 6.1):

Type I - Self aligning spherical, ring or disc lubricated.
Type II - Self aligning pivot mount, ring or disc lubricated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS
FEDERAL

WW-P-471 Pipe-Fittings (Bushings, Plugs, and Locknuts), Bronze and Ferrous (Screwed).

MILITARY

MIL-P-116 Preservation, Methods of.
MIL-S-890 Steel: Forgings and Bars for Hulls, Engines and Ordnance (Heat Treated).
MIL-P-17286 Preservation, Packaging, Packing and Marking of Propulsion and Auxiliary Steam Turbines and Gears, (Includes Associated Repair Parts).

STANDARDS
MILITARY

MIL-STD-10 Surfaces, Roughness, Waviness, and Lay.

MIL-STD-278 Welding and Allied Processes for Machinery for Ships of the United States Navy.

PUBLICATIONS
Bureau of Ships Manual
Chapter 45 - Lubrication systems.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.).

SOCIETY OF AUTOMOTIVE ENGINEERS

Standards for Pins, Cotter (Split).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. 29 West 39th Street, New York 18, New York).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING MATERIALS

A27 - Mild-to Medium Strength Carbon, Steel Castings for General Application.
A53 - Welded, and Seamless Steel Pipe.
A107 - Hot Rolled Carbon Steel Bars.
A108 - Cold-Finished Carbon Steel Bars and Shafting.
A131 - Structural Steel for Ships.
A148 - High-Strength Steel Castings for Structural Purposes.
A235 - Carbon Steel Forgings for General Industral Use.
A365 - Cold-Rolled Carbon Steel Sheets, Special Killed for Miscellaneous Drawn.
A373 - Structural Steel for Welding.
A414 - Carbon Steel Sheets of Flange and Firebox Qualities.
B21 - Naval Brass Rod, Bar and Shape.
B23 - White Metal Bearing Alloys.
B62 - Composition Brass or Ounce Metal Castings.
B143 - Tin Bronze and Leaded Tin Bronze Sand Castings.
B145 - Leaded Red Brass and Leaded Semi-Brass Sand Castings.
B271 - Copper-Base Alloy Centrifugal Castings.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania).

3. REQUIREMENTS

3.1 Definitions. - Unless otherwise specified hereinafter, terms and expressions shall be interpreted in accordance with definitions contained in Webster's New International Unabridged Dictionary.

3.1.1 Propulsion line shaft bearing. - Propulsion line shaft bearings are defined as a self-aligning bearing spaced at intervals along a line of shafting located inside the ship between the aftermost propulsion gear, engine or motor flange and the stern tube seal. The bearing is to be supported by the inner foundations of the ship and shall be oil lubricated.

3.1.2 Bearing pedestal. - Bearing pedestal is defined as the bottom part of the bearing housing, which positions and aligns the bearing shell. The pedestal supports the entire assembly of the bearing and is bolted to the bearing foundation.

3.1.3 Bearing cap. - Bearing cap is defined as a detachable top bearing housing part whose primary function is to secure bearing elements within the bearing housing. The bearing cap is bolted to the bearing pedestal.

3.1.4 Bearing housing. - Bearing housing is defined as the assembly of the bearing pedestal and cap.

3.1.5 Bearing shell. - Bearing shell is defined as the self-aligning detachable bearing parts, consisting of top and bottom segments whose inside diameter is cylindrically machined to receive a precision insert or, in the case of a thick walled bearing, is cylindrically machined to be babbitted.

3.1.6 Oil disc. - An oil disc is defined as an annulus of thin metal clamped to a basically horizontal shaft journal in such fashion as to have its lowest segment immersed in an oil-filled sump. Upon rotation of the shaft journal, the attached oil disc will carry oil adhering to the immersed portion of disc above the journal, where the adhering oil can be wiped off the disc and diverted to the bearing.

3.1.7 Oil ring. - An oil ring is defined as a fabricated metal ring of approximately triangular cross-section, whose inside diameter is approximately 1-1/3 the shaft journal diameter. The oil ring is hung on a basically horizontal shaft journal so as to rotate with the shaft. The lower segment of the ring shall be immersed in and drawn through an oil-filled sump. Oil adhering to the ring will be carried to the top of the shaft for lubrication of the bearing.

3.1.8 Bearing insert. - A bearing insert is defined as a basically cylindrical split bearing made to tolerances and finishes which eliminate the necessity for machine and hand tool fitting to a shaft journal upon installation.

3.2 Materials. - Unless otherwise approved by the bureau or agency concerned, materials shall be in accordance with the applicable documents specified in table I, and shall be used in the construction of the line shaft bearing components.

Table I Materials

Components	Applicable documents
Anti-friction metal	Grade 2 or 3 of ASTM Publication B23
Bearing precision insert backing	ASTM Publication B143, alloy 1B or ASTM Publication B271
Bolts, nuts, studs and tap rivets	Specification MIL-S-890, Handbook H-28, ASTM Publication A148

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

Table I Materials (Continued)

Components	Applicable documents
Bearing shell	ASTM Publication A27, grades 60-30; ASTM Publication A131, grade C; ASTM Publication B143, alloy 1B; or ASTM Publication B271
Bearing pedestals and caps	ASTM Publication A27, grade 60-30; ASTM Publication A131, grade C; ASTM Publication A148, all grades
Chock	ASTM Publication A414, ASTM Publication A365
Cotter pins, Oil gage, pipe brackets and clips	SAE Standards, ASTM Publication A373, ASTM Publication A235, ASTM Publication A535
Oil ring or disc and insert pieces	ASTM Publication B21, ASTM Publication B62, ASTM Publication B145
Hand hole covers, sight hole covers	ASTM Publication A27, grade 60-30; ASTM Publication A131, grade C; ASTM Publication A148, ASTM Publication A107; or ASTM Publication A108
Oil level indicator and associated equipment	Specification WW-P-471

3.2.1 Except where specifically approved by the bureau or agency concerned, the use of common cast iron parts will not be permitted. All materials used in the bearing assembly shall be compatible with respect to corrosion when exposed to a sea atmosphere and the lubricating oil selected for the unit.

3.2.2 The use of critical and strategic materials, as defined by the bureau or agency concerned, shall be kept to a minimum.

3.2.3 The use of materials differing from those specified herein will be considered bv the bureau or agency concerned, when

(a) the bearing manufacturer shows the necessity for such substitutions;
(b) materials proposed are satisfactory substitutes, and
(c) the materials are readily available.

The use of materials not specified herein does not waive the requirements that the materials shall receive the usual thorough inspection required by specifications for materials.

3.3 Equipment, fittings and accessories to be furnished. - Equipment, fittings, and accessories shall be furnished with the bearings as follows:

(a) Identification plates.
(b) Oil level indicator and related equipment.
(c) Onboard repair parts.
(d) Special tools and equipment.
(e) Bearing lifting gear, (such as chain falls and hydraulic jacks if those on the ship do not have the load capability required.)

3.4 Design. -

3.4.1 General design. - The propulsion line shaft bearings shall be constructed to assure reliability, and no compromise of this principle shall be made with any other basic design requirement. The bearing shall consist of a self-aligning bearing shell, with housing constructed to facilitate removal of, or replacement of essential components. The bearings shall have a self contained lubrication system. The housing shall consist of a pedestal and cap constructed of either fully annealed, welded or cast steel. All bearings shall be either ring or disc oiled. The design shall utilize a thin precision insert type babbitted bearing liner or a thick-walled babbitted shell, mounted in the afore described self-aligning mount. Bearing assemblies shall be designed so that sub-assemblies and parts shall be interchangeable (for any one stock number). Bearings shall be designed to withstand the shock loading of 16G's in the horizontal and 25G's in the vertical plane or as shown on figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

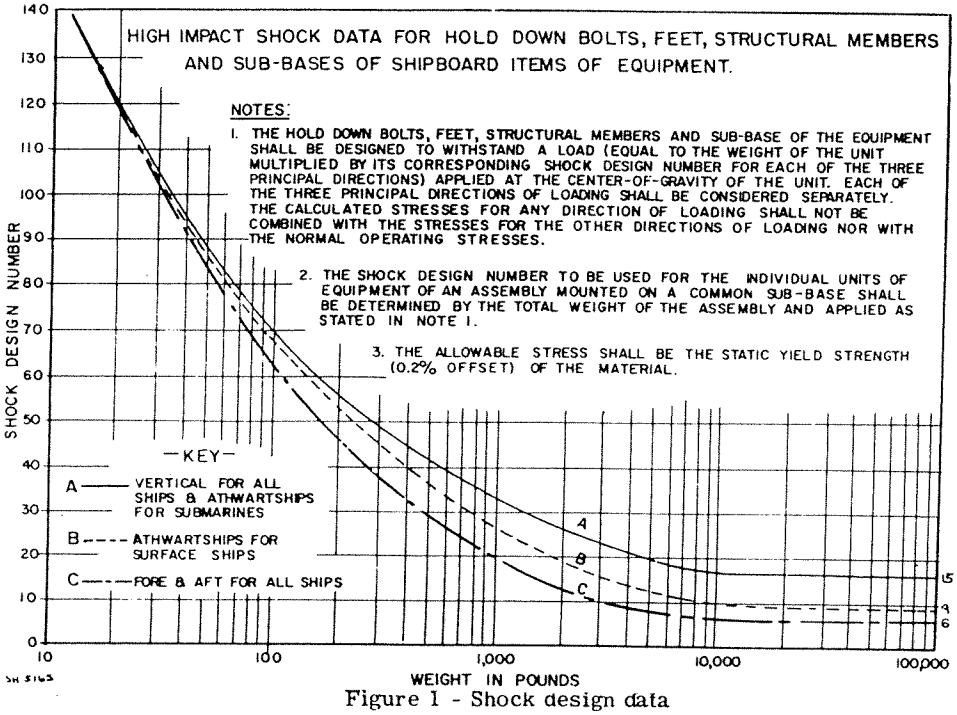


Figure 1 - Shock design data

3.4.2 Bearing housing. - The housing shall be in halves, consisting of a pedestal and cap. The bearing housing shall be designed to permit the oil rings to rotate freely up to a maximum roll of 30 degrees and a maximum pitch of 5 degrees or as specified (see 6.1). The oil rings shall not rub or strike against the side or ends of the housing oil reservoir, and shall not show a pronounced irregularity of movement. Disc lubrication shall be used for submarines and where pitch and roll requirements exceed the foregoing. The housing design shall permit the bottom bearing precision insert and shell to be freely rolled out without removal of the shaft, when the weight of the shaft is taken by lifting gear. The lifting gear for this purpose shall consist, in general, of a yoke of ample strength supported by suitable jacking bolts. A means shall be provided to prevent the rings from being thrown out of place when the ship is permanently listed, rolling, or when subjected to shock. The housing shall be designed to prevent the escape of oil along the shaft. This shall be accomplished by the use of slingers, baffles and drips. An oil trough shall be provided in the pedestal base with suitable provision for the drainage of same applied. The pedestal shall be provided with hand hole, inspection and cleanout removable plates to permit cleaning of and access to all bearing housing cavities. The pedestal shall have a faced flange, secured to the ship's foundation by forged steel bolts which will be provided by the shipbuilder. The bearing cap shall provide an opening for inspection of the oil rings and bearing. The use of felt and friction type seals is not acceptable.

3.4.3 Oil rings. - The number of oil rings shall be such that no ring is required to distribute lubricant for an axial distance greater than 7 inches on either side. The ratio of the inside diameter of the ring to the diameter of the shaft shall be in the range of 1.5 to 1.25, decreasing with increased shaft diameter. The oil ring cross section shall be triangular and designed to conform with the demand and lubricant peculiar to the individual application and shall have grooves to break the oil film. The oil ring shall maintain at least 1-inch submersion in the oil (see 6.1) reservoir under extreme conditions of roll, pitch and list, as specified (see 6.1). The oil rings shall be constructed from a single piece but of a split design to facilitate removal or replacement. (The rings shall be locked together by a suitable locking device.)

3.4.4 Oil disc. - Oil discs shall be firmly clamped to the shaft journal. Provision shall be made for removal or replacement. Oil discs shall be provided with scrapers at the top or other location to give the most effective oil flow from the disc. The oil scrapers shall also divert the oil to the bearing lining.

3.4.5 Bearing shell. - The bearing shell shall be made in halves and mounted in their respective pedestal and cap. The bearing shell spherical and pivot type self-aligning mount shall be designed to allow for thermal expansion while in operation, or operating under extreme conditions. In the case of precision inserts, the surface of the bearing shell shall be machined to receive and lock the precision inserts, which shall be designed to provide the proper force

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

fit. The bearing shell shall permit the insertion and removal of precision insert bearings without difficulty. Thick-walled bearing shells shall permit easy insertion and removal in the housing.

3.4.5.1 Bearing shell precision insert. - The insert or shell shall be preheated and tinned before lining with anti-friction metal. The bearing anti-friction metal shall have a tin base. In the case of the precision type, the babbitt thickness should not exceed 3/32 inch. The babbitted precision inserts shall be secured to the inside diameter of the spherically backed shells by screws and the top halves shall be doweled to the lower. The outside diameter shall be accurately machined with a free spread of from 0.002 to 0.006 inches to insure a force fit in the bearing shell. The maximum recommended loading on ring lubricated bearings in 50 psi with a maximum length to diameter ratio of 2. The precision insert wall thickness shall be as follows:

Shaft diameter Inches	Insert wall thickness Inches
12	7/8
16	1-1/8
20	1-1/4
24	1-3/8
28 and over	1-1/2

For bearings with two or more oil rings, the upper half precision insert can be omitted between oil rings; two pieces are required, one at each end of the bearing. The backing structure of the shell shall be steel or bronze.

3.4.5.2 Anti-friction surfaces. - On thick-walled bearing shells, the babbitt thickness shall be as follows:

Shaft diameter Inches	Babbitt thickness Inches
12	1/16
24	3/32
32	1/8
Over 32	3/16.

Grooves and holes for bonding will be permitted but are not required. All bearing surfaces shall be of anti-friction metal, grade 2 or 3 of ASTM Publication B23 and shall be metallurgically (tin) bonded to the bearing insert or shell by pouring under a definite head, if statically cast, or by centrifugal casting. Bearing surfaces shall be smoothly and accurately (32 rms) finished to designed clearances. The anti-friction metal shall be slightly relieved along the joint seam, except for a narrow strip at the ends, where the full amount of metal shall be retained to prevent loss of oil (except that the provision shall be made for a small chamfer at the joint to prevent scoring of the journal). No grooving will be permitted in the load-carrying surface of a cylindrical bearing, except for circumferential drain grooves at one or both ends of the bearing. Bearings of special design will be given consideration. (Grooving, relieving or special geometrical contours will be allowed in non-load carrying surfaces.)

3.4.6 Threaded fastenings. - Threads shall conform to Handbook H28.

3.4.7 Welding, brazing and allied processes. - Welding, brazing and allied processes shall conform to Standard MIL-STD-278. Prior to assembly of finished bearings, all cast or welded bearing parts shall be thoroughly cleaned of sand, weld spatter, or other foreign matter such as machining chips.

3.4.8 Clearance requirements. - All radial clearances shall conform to those specified in table II.

Table II - Line shaft spring bearing clearances

Journal diameter	Minimum clearances	Maximum clearance	Clearance at which bearing should be rebabbitted
Inches	Inch	Inch	Inch
1	0.007	0.009	0.017
2	.007	.009	.017
3	.007	.009	.017
4	.007	.009	.017
5	.008	.010	.019
6	.010	.012	.022
7	.011	.013	.025
8	.012	.014	.028
9	.013	.015	.030
10	.015	.018	.033
11	.016	.019	.036
12	.017	.020	.038
13	.018	.021	.040
14	.020	.023	.042

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

Table II - Line shaft spring bearing clearances - Continued

Journal diameter	Minimum clearances	Maximum clearance	Clearance at which bearing should be rebabbitted
Inches	Inch	Inch	Inch
15	.021	.025	.044
16	.022	.026	.046
17	.023	.027	.047
18	.024	.028	.049
19	.025	.029	.051
20	.027	.032	.052
21	.028	.033	.054
22	.029	.034	.055
23	.030	.035	.056
24	.031	.037	.058
25	.032	.038	.059
26	.033	.039	.060
27	.034	.041	.061
28	.035	.042	.062
29	.036	.043	.063
30	.037	.045	.064

3.4.9 Lubrication system.- Lubrication systems shall be in accordance with Bureau of Ships Manual, chapter 45. Lubrication systems shall be remote and independent of the ship's main lubricating system. The system shall be self-contained and capable of delivering the required volume of oil by utilizing either a ring or disc oiling mechanism to maintain a constant oil film on the bearing. The oils selected for the unit shall be compatible with respect to corrosion of all the assembly materials.

3.5 Repair parts and stock repair parts.-

3.5.1 Repair parts are of two general categories; namely onboard repair parts and stock repair parts. Ships Parts Control Center (SPCC), Naval Supply Depot, Mechanicsburg, Pa. is the supply demand control point (SDCP) for all repair parts.

3.5.1.1 Onboard repair parts.- Onboard repair parts are those bearing parts which are required to be carried onboard Naval ships for routine maintenance of parent equipment by shipboard personnel. These parts are listed in the ship's onboard allowance and can be installed by the ship's crew. Onboard repair parts shall be furnished and concurrently produced with line shaft bearings specified herein.

3.5.1.2 Onboard repair parts shall be furnished by the manufacturer with the original equipment and shall consist of precision inserts or babbitted shells. The quantity of onboard repair parts shall be: One insert set or babbitted shell set for each size bearing per line of shafting. These inserts or babbitted shells shall be interchangeable for a given size bearing.

3.5.1.3 Stock repair parts.- Stock repair parts are those bearing parts which are considered essential to replenish the onboard usage of repair parts for a period of one year, plus such additional parts that are subject to such infrequent breakdown and slight wear that they do not qualify for inclusion as onboard repair parts but should be immediately available in the supply system for low-rate usage replacement.

3.6 Special tools and equipment.- Special tools and equipment shall be supplied as specified hereinafter, and shall be carried onboard. Special tools are defined as those tools not listed in the Navy Stock List of General Stores (Copies of this stock list may be consulted in the office of the Government inspector.)

3.6.1 Any special tools needed for removal and replacement of line shaft bearing precision inserts and shells shall be furnished.

3.6.2 One complete set of lifting and handling gear for removal and replacement of precision bearing inserts or shells shall be furnished. This lifting gear shall be capable of lifting that portion of the shaft weight supported by the line shaft bearing.

3.7 Drawings and bill of material.- An assembly drawing and bill of material shall be required.

3.7.1 Assembly drawings.- Assembly drawings shall include the names of all parts shown therein, their respective weights, lube oil capacity and requirements. Bearing rating load, speeds and friction loss shall also be shown on the assembly drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

3.7.2 Unless otherwise specified herein, drawings shall specify surface finishes in conformance with Standard MIL-STD-10.

3.7.3 If drawing approval action is taken by an activity other than the bureau or agency concerned, a copy of that activity's approval letter shall be forwarded to the bureau or agency concerned along with two prints of each approved drawing.

3.8 Identification plates.- Each assembly shall be marked and identified by means of an identification plate. Each identification plate shall be made from rolled brass of commercial composition. The required data shall be stamped on the identification plate in 1/4 inch high capital block letters. The identification plate shall be securely fastened to the pedestal of each bearing assembly with machine screws or hardened steel drive screws. Each identification plate shall contain the following information:

Line shaft bearing (location) _____.
Ship class _____.
Federal stock number _____.
Drawing number _____.
Contract number _____.
Manufactured by _____.
Date manufactured _____.

3.8.1 Federal stock number.- The applicable Federal stock number shall be included in the identification plate data. The contractor shall obtain the applicable stock number from the supply demand control point for inclusion in the identification plate data prior to shipping. If impractical to have an identification plate attached to material, the stock number should be stamped or imprinted on material in some acceptable manner.

3.9 Workmanship.- All workmanship shall be executed in a thorough workmanlike and substantial manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.-

4.2.1 Lot.-

4.2.1.1 Assembled bearings and removable inserts.- All assembled bearings and removable inserts of the same type and size presented for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.1.1.1 Repair parts.- All repair parts of the same type and size presented for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for visual and dimensional examination.- All assembled bearings and removable inserts in the lot shall be examined as specified in 4.3.

4.2.3 Sampling for tests.- From each lot of assembled bearings and removable inserts, a sample shall be selected for the tests specified in 4.4. The babbitt and backing material will be of the same heat of metal as the bearings. The sample may be flat or curved. The babbitt and backing material shall be of approximately the same thickness as the bearing. The sample shall be at least 2 inches square.

4.2.3.1 If more than 25 percent of the area of the babbitted sample fails to pass the babbitt bond test specified in 4.4.1, or if the mean bond strength is below 8000 psi, the entire lot shall be rejected until such time as the contractor demonstrates that he has improved his procedures so as to produce an acceptable bond.

4.3 Visual and dimensional examination.- Each assembled bearing and removable insert shall be visually and dimensionally examined in accordance with the applicable drawings and the requirements of this specification not involving tests to verify conformance with the requirements of this specification. Each assembled bearing and removable insert shall be individually accepted or rejected.

4.4 Tests.- The sample assembled bearing and removable inserts selected in accordance with 4.2.3 shall be subjected to the destructive babbitt bond test performed by the Chalmers or tension method (see 4.4.1 and 4.4.2) to determine ductility and bond strength. This shall exceed a mean value of 8000 psi.

4.4.1 Chalmers bond test method.- The strength of the bond between the babbitt and backing material may be determined by the Chalmers bond test method in which an annular area of bond is isolated and forced off the bearing shell as shown on figure 2. This method requires babbitt thicker than 0.15 inch and may be performed in the unmachined bearing bore. The annular area bond is isolated by drilling through the babbitt with a trepanning drill (OD = d_3, ID = 0.350 inch) and drilling a co-axial hole in the backing

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

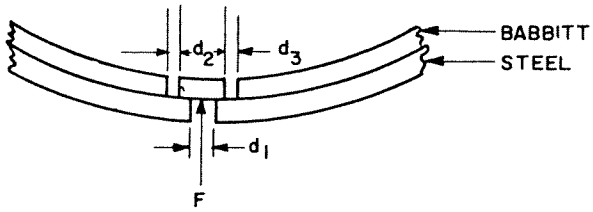


Figure 2 - Chalmers bond test method

material with a flat-ended drill (OD = d_1 = 0.250 inch). The babbitt plug is then forced off the backing material. The force in pounds required to push the plug from the shell is measured (pneumatically or hydraulically) and the bond strength in pounds per square inch is calculated by dividing this force by the area of the annulus between the hole in the babbitt and the hole in the backing material.

Bond strength = $S = F/A = F/\frac{\pi}{4}(d_2^2 - d_1^2)$

Where:

F = Rupture force, pounds.
d_2 = ID of trepanning drill, inches.
d_1 = OD of flat-end drill, inches.

4.4.2 Tensile bond test method.- The strength of the bond between the babbitt and the backing material may be determined by this method in which a rectangular area of bond (0.050 x 1 inch) is isolated and sheared off the back as shown on figure 3.

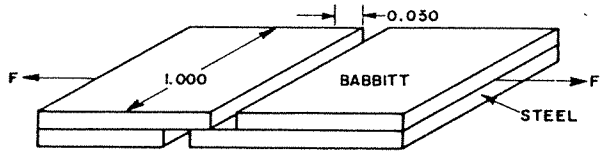


Figure 3 - Tensile bond test method

The rectangular area of bond is isolated by a saw cut through the babbitt on one side and through the backing material on the other side. The sample is made one inch wide and any desired length and thickness with the saw cuts 0.050 inches apart. The sample is pulled in a tension machine causing the bond to shear. The bond strength is equal to the force divided by the bond area. (Bond strength = S = F/A = F/0.05 x 1 inch^2). The bond strength measured by this method should be of the same order of magnitude as the Chalmers test method (see 4.4.1). If the bond shears at the babbitt-backing material interface, by this or the preceding method, it is brittle and if it shears in the babbitt it is a ductile bond. The bond should be ductile, not brittle. The rate of load application in either case should be 0.35 inch per minute at ambient temperature.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early installation and for storage of onboard repair parts and tools.-

5.1.1 Bearing assemblies.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the using activity and early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical damage and mechanical damage during direct shipment from the physical source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include Federal stock number, nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts and tools.- Onboard repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with Specification MIL-P-17286. Items not covered therein shall be unit protected and packaged in accordance with guide lines of Specification MIL-P-116. Items shall be individually packaged except when used in sets.

5.2 Domestic shipment and storage or overseas shipment requirements.- The requirements and levels of preservation packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.)

5.2.1.1 Preservation and packaging.- Bearing assemblies shall be preserved and packaged in accordance with Specification MIL-P-2845.

5.2.1.2 Repair parts and tools.- Repair parts and tools shall be preserved and packaged level A in accordance with Specification MIL-P-17286.

5.2.1.3 Packing.-

5.2.1.3.1 Bearing assemblies.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

5.2.1.3.1.1 Level A.- Bearing assemblies shall be packed for shipment in accordance with the overseas requirements of Specification MIL-P-2845.

5.2.1.3.1.2 Level B.- Bearing assemblies shall be packed for shipment in accordance with the domestic requirements of Specification MIL-P-2845.

5.2.1.3.2 Repair parts and tools.- Repair parts and tools shall be packed level A or B as specified in accordance with Specification MIL-P-17286.

5.2.1.4 Marking.- In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable referenced packaging specification and Standard MIL-STD-129.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Maximum degrees of roll and pitch (see 3.4.2).
(d) Submersion in the oil of the oil ring and list (see 3.4.3).
(e) Preservation, packaging, packing and marking requirements other than those required by 5.1 (see 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy-Bureau of Ships
(Project 2010-N017Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.